Echo Yeast hardware module — final purchase list

Updated 2026-10-14 · Items already ordered and newly ordered · Unit prices are the amounts actually paid

1. Itemised purchase list

No.	Category	Item	Specification / model	Qty	Unit price (CNY)	Subtotal (CNY)	Purpose
1	Structural parts & printing consumables	3D printing filament (PLA)	PLA Basic, white 1000 g + black 1000 g (without spool)	1 spool each	77.38	77.38	FDM printing of the collection cup, the heater-block body and the three plug-in modules
2	Temperature control	Heat-resistant thermal silicone pad	200 × 400 × 0.3 mm, 6 W/(m · K)	1 sheet	25	25	Thermal interface between the heating film and the heater block; cut to size
3	Temperature control	PI heating film	60 × 70 mm, 12 V / 25 W	2 pcs	24.64	24.64	Bonded to the underside of the heater block as the heating element
4	Temperature control	MAX31865 module + PT100	MAX31865 converter module + PT100 platinum RTD (3-wire)	1 set	33.5	33.5	Temperature acquisition: PT100 resistance converted to a digital temperature by a 16-bit ADC
5	Temperature control	KSD9700 thermal cut-off	60 ° C, 250 V / 10 A, metal case, normally closed	1 pc	6.12	6.12	Hardware-level over-temperature protection; in series with the heater so it physically cuts the circuit
6	Temperature control	MOSFET driver module	IRF520	1 pc	3.29	3.29	Takes the PWM signal and drives the heating film to implement PID power control
7	Temperature control	Cooling fan	40 × 40 × 10 mm, 12 V, quiet	1 pc	10	10	Mounted on the heat sink to keep the electronics underneath and the block from overheating
8	Temperature control	Heat sink	Aluminium, 70 × 36 × 100 mm	1 pc	23	23	Dissipates heat downwards and protects the circuit underneath
9	Optics	LED light source	485 nm high-brightness LED	3 pcs	17.34	17.34	Excitation light source; spares kept for wavelength margin
10	Optics	LED constant-current driver	Adjustable constant-current driver board, 350 / 500 mA	1 pc	36.38	36.38	Provides a stable current for the 485 nm LED and reduces intensity drift
11	Optics	Photodiode	BPW34, through-hole package	4 pcs	8.1	8.1	Detects the 528 nm fluorescence at the receiver end; spares kept
12	Optics	Narrow-band filters	Ø10 mm, 30 nm bandwidth, OD3; 485 nm × 2 + 528 nm × 2	4 pcs	154	154	One at each end separates the excitation light from the fluorescence spectrum
13	Optics	Op-amp	OPA350UA (SOIC-8)	2 pcs	18.8	18.8	Forms the transimpedance amplifier with the photodiode and outputs a measurable voltage
14	Optics	Light shield (black heat-shrink tubing)	12 mm bore, 10× thick wall	1 pack	8	8	Wraps the emitter and receiver plugs to exclude ambient light
15	Optics	Focusing lenses for emitter and receiver	Matched to the 485 nm emitter and the 528 nm receiver (newly ordered)	2 sets	TBD	TBD	Focus the LED output onto the centre of the vessel and the fluorescence onto the photodiode active area

No.	Category	Item	Specification / model	Qty	Unit price (CNY)	Subtotal (CNY)	Purpose
16	Electronics & signal	Microcontroller	Arduino Nano V4 (board + USB cable)	1 pc	62.84	62.84	Timing control, ADC readout and the PID temperature algorithm
17	Electronics & signal	ADS1115 high-precision ADC	ADS1115 module, 16-bit, 4-channel differential, I ² C (newly ordered)	1 pc	TBD	TBD	Reads the amplified photodiode voltage with high precision
18	Electronics & signal	SOT89 adapter board	SOT89 to DIP through-hole adapter (newly ordered)	TBD	TBD	TBD	Lets SOT89 SMD parts be soldered through-hole
19	Electronics & signal	SOT223 adapter board	SOT223 to DIP through-hole adapter (newly ordered)	TBD	TBD	TBD	Lets SOT223 SMD parts be soldered through-hole
20	Power & wiring	Power adapter	24 V / 2 A (48 W), with mains lead	1 pc	12.87	12.87	Supplies the instrument; 24 V bus
21	Power & wiring	LM2596 buck module	LM2596S DC-DC adjustable step-down	1 pc	4.49	4.49	Steps 24 V down to 12 V (heating film, fan) and 5 V (logic)
22	Power & wiring	Breadboard / protoboard	Single-sided, rosin-coated green, 12 × 18 cm	2 pcs	10.87	10.87	Prototyping during debugging, then soldered permanently
23	Power & wiring	Jumper wires	21 cm, male – male / male – female / female – female, 120 pcs total	1 set	9.87	9.87	Temporary connections between boards and modules
24	Power & wiring	Electrolytic capacitors	16 V / 100 μ F, 5 × 11 mm	2 pcs	6.74	6.74	Supply filtering; suppresses switching noise
25	Power & wiring	Resistor kit	1/4 W metal film, 25 common values, 500 pcs total	1 set	7.9	7.9	Current limiting, voltage division and transimpedance feedback
26	Power & wiring	Indicator LEDs	5 mm red / green / yellow, 60 pcs total	1 set	3.46	3.46	Power and run-status indication
27	Power & wiring	Terminal blocks	KF301-2P × 10 + 3P × 10, 5.08 mm	1 pack each	7.17	7.17	Power input and load wiring
28	Power & wiring	30 AWG silicone wire	Extra-flexible silicone wire for wiring inside the plugs and for signals (newly ordered)	TBD	TBD	TBD	Fine wiring inside the plugs keeps them compact and flexible
29	Tools & consumables	Thermal silicone adhesive	Kafuter K-5204K, fast-curing, white	1 tube	13.2	13.2	Fixes the heating film, PT100 and heat sink; also seals against light
30	Tools & consumables	Heat-resistant double-sided tape	PI polyimide tape, 0.1 mm thick, 3 mm × 10 m	1 roll	10.7	10.7	Holds the heating film onto the heater-block surface
31	Tools & consumables	Silicone wire	22 AWG red + black, rated ≥ 105 ° C	1 roll each	13.05	13.05	Power wiring for the heater and supply circuits
32	Tools & consumables	Desoldering wick	For desoldering and rework (newly ordered)	1 roll	TBD	TBD	Rework and clearing of mis-soldered joints
33	Tools & consumables	Multimeter	UNI-T UT33A, auto-ranging	1 pc	73	73	Voltage, current, continuity and resistance measurement

No.	Category	Item	Specification / model	Qty	Unit price (CNY)	Subtotal (CNY)	Purpose
34	Tools & consumables	Soldering station (936)	60 W adjustable constant temperature, standard model	1 set	66	66	Soldering the circuits and plug-in modules
35	Tools & consumables	Hot-melt glue gun	40 W, with glue sticks	1 pc	15.35	15.35	Securing components and sealing light leaks
36	Tools & consumables	Solder wire	Lead-free, 63% tin, 0.8 mm, rosin core	1 roll	27.65	27.65	Soldering consumable
		Total, items already ordered (CNY)				790.71	

2. Summary by category

Summary by category (items already ordered)	None	None	None
Category	Items	Amount (CNY)	Share
Structural parts & printing consumables	1	77.38	9.79%
Temperature control	7	125.55	15.88%
Optics	7	242.62	30.68%
Electronics & signal	4	62.84	7.95%
Power & wiring	9	63.37	8.01%
Tools & consumables	8	218.95	27.69%
Total	36	790.71	100.00%

Unit prices are the amounts actually paid; amounts are in CNY.

3. Purchase links (items already ordered)

Item	Link
Item	Link
3D printing filament (PLA)	https://e.tb.cn/h.8D2q9Wgd9FICQlA?tk=4wliTOpQjq4
Heat-resistant thermal silicone pad	https://e.tb.cn/h.8D2u35e03HzrnKQ?tk=ZUlpTOpnJNj
PI heating film	https://e.tb.cn/h.8DdYQSZ9I7Gn80H?tk=q5L3TOpvEUn
MAX31865 module + PT100	https://e.tb.cn/h.8Eo6m9Y1yBcqsCB?tk=0hdiTOpzeHk
KSD9700 thermal cut-off	https://e.tb.cn/h.8EoisstnhcDy4WH?tk=nJM9TOKZxsu

Item	Link
MOSFET driver module	https://e.tb.cn/h.8FTxa7A424Jc4TE?tk=Gf2cTOKc66T
Cooling fan	https://e.tb.cn/h.8FTCMdiqBypUWOx?tk=15evTOKX21L
Heat sink	https://e.tb.cn/h.8vh4852tGrpl6rI?tk=MU7NTOKf3Em
LED light source	https://e.tb.cn/h.8vhUy8oFTunUWog?tk=3NKaTOKg7rV
LED constant-current driver	https://e.tb.cn/h.8EoqXI9Ckjpxpg9?tk=B6LBTOK6QOH
Photodiode	https://e.tb.cn/h.8vhSus9evlimDIM?tk=drwcTOK7u8T
Narrow-band filters	https://e.tb.cn/h.8F6ngurIWsopc7N?tk=susnTOKFY6Y
Op-amp	https://e.tb.cn/h.8DWYYiqtEKvqeLx?tk=O7y0TOKxbDm
Light shield (black heat-shrink tubing)	https://e.tb.cn/h.8ELgpJKO9UjsSkU?tk=7ORATOKzjE0
Microcontroller	https://e.tb.cn/h.8ELBloVll81AYYP?tk=ZfVFTOqJpe2
Power adapter	https://e.tb.cn/h.8F6tQHNABFojNUa?tk=Vds9TOKARZj
LM2596 buck module	https://e.tb.cn/h.8DWdMqYnWoZeTiV?tk=qyQATOqZxlt
Breadboard / protoboard	https://e.tb.cn/h.8DWeeX1yDMySMdW?tk=gWEGTOqbyTD
Jumper wires	https://e.tb.cn/h.8vS0UJczgCS6B9I?tk=UVU1TOqX7Bc
Electrolytic capacitors	https://e.tb.cn/h.8DWTRQb5qqQUXFB?tk=XnXFTOq3l20
Resistor kit	https://e.tb.cn/h.8Fh0TDbu7fD8gpo?tk=QrEITOq4cHV
Indicator LEDs	https://e.tb.cn/h.8ELnMVRucfBtMhc?tk=ielnTOqUEXm
Terminal blocks	https://e.tb.cn/h.8ELo5JbhvB3KreD?tk=uO7KTOqgam2
Thermal silicone adhesive	https://e.tb.cn/h.8Fhd5G7Prf3J7S0?tk=t5SDTOqT7Sk
Heat-resistant double-sided tape	https://e.tb.cn/h.8DWQ2ns5GLIGnhN?tk=KnPSTOqSk57
Silicone wire	https://e.tb.cn/h.8ELJRdCyEZp1FDH?tk=5qyfTOq7oBz
Multimeter	https://e.tb.cn/h.8Fh5C8Nsg2KiLLM?tk=q5DbTOqjBmh
Soldering station (936)	https://e.tb.cn/h.8DWpiXAZNefdTRU?tk=A0vmTOqObkV
Hot-melt glue gun	https://e.tb.cn/h.8ELFziG8HafPdXK?tk=Mr0HTOqNcMG
Solder wire	https://e.tb.cn/h.8FhLCVc3yUmjPZa?tk=Lg8xTOquJTR